seek.onlinedegree.iitm.ac.in/courses/ns_25t2_cs2101/outline?id=36
Screen Capture Just now
Screenshot captured
You can paste the image from the clipboard

- **(d) Noise in the input signal**: Noise in the input signal can degrade the accuracy of ADC conversions by introducing errors or uncertainties in the measured signal.

Therefore, the correct answer is **(c)** because the type of microcontroller used does not directly impact the accuracy of ADC conversions; it primarily affects other aspects such as processing power, peripherals, and features unrelated to ADC accuracy.

Accepted Answers:

The type of microcontroller used

10) What is the primary purpose of using PWM in microcontroller applications? *2 points*

- To generate high-frequency signals for communication
- To control the brightness of LEDs and speed of motors
- To convert digital signals to analog signals
- To store data in memory

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

Explanation:

- **(a) To generate high-frequency signals for communication**: While PWM can generate signals, its primary application in microcontrollers is not for high-frequency communication signals but rather for controlling the duty cycle to control devices like motors and LEDs.
- **(b) To control the brightness of LEDs and speed of motors**: This is correct. PWM is widely used to vary the average power delivered to devices like LEDs (for brightness control) and motors (for speed control) by adjusting the duty cycle of the PWM signal.
- **(c) To convert digital signals to analog signals**: This function is performed by a Digital-to-Analog Converter (DAC), not PWM.
- **(d) To store data in memory**: Storing data in memory is unrelated to the function of PWM.

Therefore, the correct and succinct answer is **(b)**.

Accepted Answers:

Feedback

- **(a) All ADCs must operate with multiple channels**: False. ADCs can operate with single or multiple channels.
- **(b) All ADCs will take multiple clock cycles to convert a voltage to a number**: True. ADCs typically require multiple clock cycles for accurate conversion.
- **(c) ADCs can only convert voltages to numbers**: False. ADCs can convert various analog signals (not just voltages) to digital form.

Since not all statements are true, the correct answer is **(e) none of the above**.

Accepted Answers:

none of the above

9) Which of the following factors does NOT affect the accuracy of an ADC conversion? *2 points*

- Sampling rate
- Resolution of the ADC
- The type of microcontroller used
- Noise in the input signal

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

Explanation:

- **(a) Sampling rate**: The sampling rate affects the accuracy of an ADC. A higher sampling rate can capture fast-changing signals more accurately.
- **(b) Resolution of the ADC**: The resolution (number of bits) determines the smallest detectable change in the analog signal. Higher resolution improves accuracy.
- **(c) The type of microcontroller used**: The type of microcontroller does not directly affect the accuracy of ADC conversions. The accuracy is primarily determined by the characteristics of the ADC itself (such as resolution, linearity, and sampling rate).

Feedback

Let's analyze each statement to understand why (e) is the correct answer:

- **(a) They must always sample at a constant rate**: This statement is false. ADCs can operate at variable sampling rates depending on the requirements of the application. Some ADCs support adjustable sampling rates, while others operate at a fixed rate but are not universally bound to a constant rate.
- **(b) They must always generate the same number of bits as output**: This statement is false. ADCs come in various resolutions, meaning they can output different numbers of bits. Common ADC resolutions include 8-bit, 10-bit, 12-bit, and higher. The resolution determines the number of discrete levels (and hence the number of bits) in the digital output.
- **(c) They are the only analog components in a uC**: This statement is false. Microcontrollers (uC) can have various analog components beyond just ADCs. They may include Digital-to-Analog Converters (DACs), analog comparators, analog filters, operational amplifiers (op-amps), and other analog peripherals depending on the specific microcontroller's design.

Given that none of the statements (a), (b), or (c) are universally true for ADCs, the correct answer is (e) none of the above. This choice accurately reflects the variability and diverse capabilities of ADCs in microcontroller systems.

Accepted Answers:

none of the above

8) Which of the following are true? *0 points*

- All ADCs must operate with multiple channels
- All ADCs will take multiple clock cycles to convert a voltage to a number
- ADCs can only convert voltages to numbers
- all of the above
- none of the above

No, the answer is incorrect.
Score: 0

Feedback:

Feedback

seek.onlinedegree.iitm.ac.in/courses/ns_25t2_cs2101/outline?id=36
Screen Capture Just now
Screenshot captured
You can paste the image from the clipboard

Feedback

A Phase-Locked Loop (PLL) is a control system that generates an output signal whose phase is related to the phase of an input signal. The primary purpose of a PLL in microcontroller and digital systems includes the following:

- **(a) Generate different clocks from a single source**: This is correct. A PLL can take a single reference clock input and multiply or divide it to generate multiple output clocks of different frequencies. This allows the system to provide various clock signals for different parts of the system, each operating at its optimal frequency.

Incorrect options:

- **(b) Take different clock sources and combine them**: Incorrect. A PLL does not combine different clock sources. Instead, it typically locks onto a single reference clock and generates a related output clock.
- **(c) Make the system run fast**: Incorrect. While a PLL can be used to increase the clock frequency, its purpose is broader than just making the system run fast. It is used to generate stable and precise clock signals at various frequencies.
- **(d) Wake the uC from deep sleep**: Incorrect. Waking a microcontroller from deep sleep is typically the role of a real-time clock (RTC) or an external interrupt, not a PLL.

In summary, the primary purpose of a PLL is to generate different clocks from a single source, making (a) the correct answer.

Accepted Answers:

generate different clocks from a single source

7) Which of the following are true for ADCs? *0 points*

- they must always sample at a constant rate
- they must always generate the same number of bits as output
- they are the only analog components in a uC
- all of the above
- none of the above

Yes, the answer is correct.
Score: 0

Feedback:

seek.onlinedegree.iitm.ac.in/courses/ns_25t2_cs2101/outline?id=36
Screen Capture Just now
Screenshot captured
You can paste the image from the clipboard

Feedback

Power consumption in an integrated circuit (IC) is influenced by several factors. Let's evaluate each option to understand why (a) and (b) are correct:

- **(a) Larger area**: A larger IC area typically means more transistors and interconnects, leading to higher capacitance. As the capacitance increases, the power required to charge and discharge these capacitances also increases, resulting in higher power consumption.
- **(b) Higher clock frequency**: Increasing the clock frequency directly increases the dynamic power consumption of the IC. Dynamic power consumption is proportional to the clock frequency ($P = CV^2f$, where P is power, C is capacitance, V is voltage, and f is frequency). Therefore, higher clock frequencies lead to higher power consumption.
- **(c) Lower voltage of operation**: Lowering the operating voltage typically reduces power consumption. This is because dynamic power consumption is proportional to the square of the voltage ($P = CV^2f$). Therefore, lower voltage leads to significantly lower power consumption, not higher.
- **(d) All of the above**: This option is incorrect because, as explained above, lower voltage of operation does not lead to more power consumption.

Given these explanations, the correct answer is:

(a) larger area and (b) higher clock frequency.

Accepted Answers:

larger area
higher clock frequency

6) Purpose of a Phase -Locked loop(PLL) is to: *2 points*

- (●) generate different clocks from a single source
- ( ) take different clock sources and combine them
- ( ) make the system run fast
- ( ) wake the uC from deep sleep

Yes, the answer is correct.
Score: 2

Feedback:

seek.onlinedegree.iitm.ac.in/courses/ns_25t2_cs2101/outline?id=36
Screen Capture Just now
Screenshot captured
You can paste the image from the clipboard

Feedback:

Feedback

In the context of a microcontroller (uC) system, high-speed peripherals are those that require rapid data transfer rates and quick response times. Let's evaluate each option to understand why none of them qualify as high-speed peripherals:

- **(a) Keyboard**: A keyboard is a low-speed peripheral. It sends signals to the microcontroller at relatively low rates since human typing speed is quite slow compared to the operational speeds of microcontrollers. Keyboards do not require fast data transfer rates.
- **(b) LCD character display**: An LCD character display typically updates at a low speed. It is designed to display characters or simple graphics, and the rate at which data is sent to update the display is relatively slow. High-speed data transfer is not necessary for its operation.
- **(c) Temperature sensor**: Temperature sensors are also low-speed peripherals. They measure and send temperature data at intervals, which does not require high-speed communication. The data rate for temperature changes is generally very slow, so these sensors operate at low speeds.

Given these explanations, it is clear that:

- **None of the above peripherals (keyboard, LCD character display, or temperature sensor) qualify as high-speed peripherals in a microcontroller system.**

Therefore, the correct answer is (e) none of the above.

Accepted Answers:

none of the above

5) Which of the following will lead to more power consumption in an integrated circuit? *2 points*

- [x] larger area
- [x] higher clock frequency
- [ ] lower voltage of operation
- [ ] all of the above

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

Running a CPU at a high clock speed can lead to several drawbacks. Here are the explanations for each option:

Correct Options:

- **(a) More power consumption**: Higher clock speeds require more electrical power because the CPU is switching transistors on and off more frequently. This increased activity demands more energy, leading to higher power consumption.
- **(c) Higher temperature**: Increased power consumption results in more heat being generated by the CPU. As the clock speed goes up, the CPU produces more heat, leading to higher temperatures. Effective cooling mechanisms are needed to manage this heat and prevent overheating.

Incorrect Options:

- **(b) More noise and vibration**: While higher power consumption can cause cooling fans to run faster and potentially create more noise, the clock speed itself does not directly cause more noise and vibration. Noise and vibration are generally more related to the cooling solutions (like fans) and mechanical components rather than the electronic operations of the CPU.
- **(d) Slow program execution**: Running at a high clock speed actually makes the CPU execute instructions faster, leading to quicker program execution, not slower. The purpose of increasing the clock speed is to improve performance and reduce the time it takes to execute programs.

In summary, running at high clock speeds leads to increased power consumption and higher temperatures, making (a) and (c) the correct drawbacks.

Accepted Answers:

more power consumption
higher temperature

4) Which of the following are likely to be high speed peripherals in a microcontroller (uC) system? *2 points*

- keyboard
- LCD character display
- temperature sensor
- all of the above
- none of the above

Yes, the answer is correct.

2) A computer working at a 1 GHz clock is always faster than a computer working at a 500 MHz clock. *2 points*

- True
- False

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

The actual speed is finally a function of how fast it can finish executing a program. Even if it runs at 1 GHz, it may take multiple cycles per instruction and take longer to finish the program.
The statement is **false** because clock speed alone does not determine a computer's performance. Other factors like CPU architecture, instructions per clock cycle (IPC), number of cores, and overall system efficiency also play crucial roles. A computer with a lower clock speed could be faster if it has a more efficient architecture or better overall system design.

Accepted Answers:

False

3) Which of the following are the drawbacks of running at high clock speed? *2 points*

- [x] more power consumption
- [ ] more noise and vibration
- [x] higher temperature
- [ ] slow program execution

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

Running a CPU at a high clock speed can lead to several drawbacks. Here are the explanations for each option:

Graded Assignment 4

1) Which of the following elements are used to store states in a computer system? *2 points*

- [x] registers
- [x] memory
- [ ] clock
- [ ] address bus

Yes, the answer is correct.
Score: 2

Feedback:

Feedback

In a computer system, the elements used to store state are:

- **Registers:** These are small, fast storage locations within the CPU that hold data temporarily during processing. Registers are crucial for the CPU to perform operations quickly.
- **Memory:** This includes both primary memory (RAM) and secondary memory (hard drives, SSDs). Memory stores data and instructions that the CPU needs to access and execute. It is a fundamental component for storing the state of running programs.

The other elements listed do not store state:

- **Clock:** This element provides the timing signals to synchronize the operations of the computer system, but it does not store any data or state.

- Address bus: This is used to transfer address information between the CPU and memory, allowing the CPU to access specific memory locations. It does not store data or state itself.

Thus, the correct answers are **registers** and **memory**.

Accepted Answers:

registers
memory

2) A computer working at a 1 GHz clock is always faster than a computer working at a 500 MHz clock. *2 points*

- ( ) True